

PHYSICS – BOARD QUESTION PAPER : JULY 2022**Time: 3 Hours****Max. Marks: 70****General Instructions:**

The question paper is divided into four sections.

1. Section A: Q1 contains Ten multiple choice questions. Q2 contains Eight very short questions.
2. Section B: Q3 to Q14 contain Twelve questions. Attempt any Eight.
3. Section C: Q15 to Q26 contain Twelve questions. Attempt any Eight.
4. Section D: Q27 to Q31 contain Five questions. Attempt any Three.
5. Use of log table is allowed. Use of calculator is not allowed.
6. Figures to the right indicate full marks.
7. For each MCQ, it is mandatory to write the correct answer along with its alphabet.
Only the first attempt will be considered.

Physical Constants:

$$\mu_0 = 4\pi \times 10^{-7} \text{ Wb/Am}$$

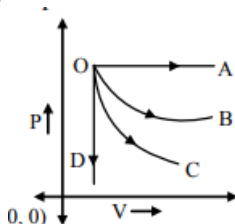
$$\sigma = 5.7 \times 10^{-8} \text{ J/m}^2 \text{ s K}^4$$

$$g = 9.8 \text{ m/s}^2$$

$$\pi = 3.142$$

SECTION – A**Q.1 Select and write the correct answers: (10M)**

- (i) Raindrops are spherical because of
- (a) surface tension (b) capillarity
(c) downward motion (d) acceleration due to gravity
- (ii) Average K.E. of a gas is
- (a) directly proportional to absolute temperature
(b) directly proportional to square of absolute temperature
(c) directly proportional to square root of absolute temperature
(d) inversely proportional to absolute temperature
- (iii) In P–V graph, curve OA represents
- (a) isochoric process
(b) isothermal process
(c) isobaric process
(d) adiabatic process



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(iv) For standing waves on string fixed at both ends

- (a) all particles vibrate in phase
- (b) all antinodes vibrate in phase
- (c) alternate antinodes vibrate in phase
- (d) all particles between two antinodes vibrate in phase

(v) Immersing diffraction apparatus in water causes

- (a) λ increases
- (b) central maximum width increases
- (c) central maximum width decreases
- (d) frequency decreases

(vi) Magnetic field at centre of circular coil varies

- (a) inversely with R^2
- (b) directly with R
- (c) inversely with R
- (d) directly with R^2

(vii) Balmer series occurs when transitions terminate on

- (a) 2nd orbit (b) 1st orbit
- (c) 3rd orbit (d) 4th orbit

(viii) SHM oscillator $A = 16$ cm, $T = 4$ s. Time from $x = 16$ cm to $x = 8$ cm is

- (a) $1/2$ s (b) $2/3$ s (c) $5/6$ s (d) $4/3$ s

(ix) In potentiometer, balance length = 240 cm. On shunting cell with $2\ \Omega$, length = 120 cm. Internal resistance is

- (a) $4\ \Omega$ (b) $2\ \Omega$ (c) $1\ \Omega$ (d) $0.5\ \Omega$

(x) Metal wire dragged in 0.5 s through flux 8×10^{-4} Wb. Induced emf is

- (a) 1.6 mV (b) 16 mV
- (c) 1.6 V (d) 16 V

Q.2 Answer the following: (8M)

(i) In which thermodynamic process does internal energy remain constant?

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(ii) State law of conservation of angular momentum.

(iii) What is shunt?

(iv) What happens to fringe width in diffraction if diameter of wire increases?

(v) What is a perfectly black body?

(vi) State formula for electric field at point outside infinitely long charged cylindrical conductor.

(vii) Half-life = 1.386 years. Calculate decay constant per year.

(viii) Engine develops 62.84 kW at 1200 rpm. What torque is delivered?

SECTION – B

Attempt any Eight questions. (16M)

Q.3. What is capillarity? State any two uses.

Q.4. Define: (i) Emissive power (ii) Coefficient of emission.

Q.5. State any two characteristics of progressive waves.

Q.6. Draw neat labelled diagram of parallel plate capacitor with dielectric slab.

Q.7. State formula for magnetic potential energy of a dipole and obtain minimum and maximum values.

Q.8. What is gyromagnetic ratio? Write expression.

Q.9. How does wave theory fail to explain photoelectric effect? (Any two points.)

Q.10. A system releases 125 kJ heat while 104 kJ work is done on it. Calculate change in internal energy.

Q.11. Plane wave $\lambda = 4000 \text{ \AA}$ on double slits. Total separation of 10 bright fringes on 2 m screen is 2 cm. Find slit separation.

Q.12. Induced emf = 96.0 mV when current in nearby coil changes at 1.20 A/s. Find mutual inductance M.

Q.13. Aircraft wingspan 50 m, $B = 6 \times 10^{-5} \text{ T}$. Find velocity needed to get 1.2 V between tips.

Q.14. Surface density of disc radius 10 cm is 2 kg/m^2 . Find MI about axis through centre, perpendicular to plane.

SECTION – C

Attempt any Eight questions. (24M)

Q.15. State zeroth law of thermodynamics. What are limitations of first law?

Q.16. What is de Broglie hypothesis? Obtain expression for de Broglie wavelength.

Q.17. Derive expression for kinetic energy of a rotating body.

Q.18. Derive laws of reflection of light using Huygens' theory.

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- Q.19. Derive orbital magnetic moment of electron and state Bohr magneton formula.
- Q.20. Explain: (a) Capacitive reactance (b) Inductive reactance (c) Impedance.
- Q.21. Define α_{dc} and β_{dc} . Obtain relation between them.
- Q.22. Open pipe has fundamental 600 Hz. First overtone of a closed pipe has same frequency as first overtone of open pipe. Find lengths of both. ($v = 330$ m/s)
- Q.23. Negative charge 1.6×10^{-19} C is in equilibrium between plates 10 cm apart with 4000 V across. Find mass of particle.
- Q.24. $L = 1000$ mH, $C = 36$ μ F, $R = 12$ Ω in series with 120 V, 50 Hz. Calculate: (i) impedance at resonance (ii) current at resonance (iii) resonant frequency.
- Q.25. Two long parallel wires, distance 2 cm, same current, force/length 4×10^{-2} N/m. Find current in each.
- Q.26. Calculate energy radiated in 30 s by black body area 200 cm² at 127°C .

SECTION – D**Attempt any Three questions. (12M)**

- Q.27. Discuss analytical composition of two linear SHMs of same period along same path. Derive resultant amplitude. Find resultant amplitude for phase difference (i) 0 (ii) $2\pi/3$ radians.
- Q.28. What is transformer? With suitable diagram, describe its working.
- Q.29. Define angle of contact. State any two properties. Find pressure difference inside/outside spherical water drop radius 2 mm ($\sigma = 73 \times 10^{-3}$ N/m).
- Q.30. State Kirchhoff's laws of electrical network. Two cells of emfs E_1 and E_2 in series aiding give balancing length 3.2 m; in series opposing give 0.7 m. Compare E_1 and E_2 .
- Q.31. State first and second postulates of Bohr's model. Compute ratio of longest wavelengths of Lyman and Balmer series in hydrogen.

